

MOHAMMADHOSEIN GHOLAMREZAEI

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[Google Scholar](#) [GitHub](#) [LinkedIn](#)

EDUCATION

Aug 2023 – Present	Ph.D. in COMPUTER SCIENCE, University of Virginia , Charlottesville, VA Advisor: PROF. KEVIN SKADRON
Sep 2021 – Aug 2023	M.E. in COMPUTER ENGINEERING, Chosun University , Gwangju, South Korea Thesis: “Hardware Acceleration of Fused-Layer CNNs via Most-Significant-Digit First Arithmetic” Advisor: PROF. JEONG-A LEE
Sep 2015 – Sep 2020	B.E. in COMPUTER ENGINEERING, Shahid Beheshti University , Tehran, Iran Thesis: “Implementation of Amateur Radio Transceiver on FPGA Platform” Advisor: DR. DARA RAHMATI

RESEARCH INTERESTS

Computer Architecture • Hardware Accelerators • Processing in Memory

RESEARCH EXPERIENCE

Aug 2023 – Present	Graduate Research Assistant , Computer Science Dept., University of Virginia Advisor: Prof. Kevin Skadron. Research on digital DRAM PIM architectures, compiler frameworks for bit-serial PIM, and hardware accelerator design.
May 2026 – Aug 2026	Research Intern , SK Hynix America, San Jose, CA
Sep 2021 – Aug 2023	Research Assistant , Computer Systems Laboratory, Chosun University Research on MSDF arithmetic-based hardware accelerators for ML inference (CNNs, SVM, kNN).
Feb 2019 – Aug 2021	Researcher , High Performance Computing Laboratory, IPM – Institute for Research in Fundamental Sciences, Tehran Research on stochastic computing, residue number systems, and FPGA-based accelerators.

SELECTED PUBLICATIONS

- [J1] F. A. Siddique, D. Guo, H. Abbot, K. Durrer, **M. Gholamrezaei**, M. Baradaran, E. Ermovick, A. Ahmed, Z. Fan, B. Gul, A. Venkat, K. Skadron, “Characterizing Digital DRAM PIM through Modeling and Benchmarking,” *ACM Transactions on Architecture and Code Optimization (TACO)*, May 2026.
- [J2] D. Guo, **M. Gholamrezaei**, M. Hofmann, A. Venkat, Z. Zhang, K. Skadron, “PIMsynth: A Unified Compiler Framework for Bit-Serial Processing-In-Memory Architectures,” *IEEE Computer Architecture Letters*, 2025.
- [C1] F. A. Siddique, D. Guo, Z. Fan, **M. Gholamrezaei** et al., “Architectural Modeling and Benchmarking for Digital DRAM PIM,” *IEEE International Symposium on Workload Characterization (IISWC)*, Aug 2024.
- [C5] M. S. Karvandi, **M. H. Gholamrezaei** et al., “HyperDbg: Reinventing Hardware-Assisted Debugging,” *ACM Conference on Computer and Communications Security (CCS)*, Nov 2022.
- [C8] S. Gorgin, **M. H. Gholamrezaei**, D. Javaheri, J.-A. Lee, “An Efficient FPGA Implementation of k-Nearest Neighbors via Online Arithmetic,” *IEEE International Symposium on FCCM*, May 2022.

- [J4] K. Givaki, A. Khonsari, **M. H. Gholamrezaei**, S. Gorgin, M. H. Najafi, “A Generalized Residue Number System Design Approach for Ultra-Low Power Arithmetic Circuits,” *IEEE Trans. Computer-Aided Design of Integrated Circuits and Systems*, March 2023.
- [J5] K. Givaki, R. Hojabr, **M. H. Gholamrezaei** et al., “High Performance Deterministic Stochastic Computing Using Residue Number System,” *IEEE Design and Test*, Jan 2021.

Full list: [Google Scholar](#)

TEACHING EXPERIENCE

<i>F2024, S2025, F2025, S2026</i>	Teaching Assistant – Computer Systems and Organization 2 (CS 3130), University of Virginia
<i>2018–2020</i>	Teaching Assistant – Microprocessors & Assembly Language (×3), Digital Design, Embedded Systems (×2), Shahid Beheshti University

HONORS & AWARDS

2021	Research Assistant Scholarship, Chosun University
2020	Candidate for Best Bachelor Thesis Award, Shahid Beheshti University
2015	Top 1%, Nationwide University Entrance Exam, Iran (180,000+ contestants)

TECHNICAL SKILLS

LANGUAGES	VHDL, SystemVerilog/Verilog, Scala/Chisel, C/C++, Python, x86/AVR/ARM Assembly, Java
TOOLS & PLATFORMS	Synopsys Design Compiler, Gem5, Xilinx Vivado/ISE, Modelsim, LLVM, Cadence Virtuoso, Git, Linux, LegUp HLS
RESEARCH AREAS	RTL Design, Hardware Accelerators, Processing-In-Memory, High-Level Synthesis, Computer Arithmetic

SELECTED PROJECTS

- **HyperDbg** – Open-source hypervisor-assisted debugger for Windows (user/kernel mode). Designed for analysis, fuzzing, and reverse engineering. github.com/hyperdbg
- **PIMsynth** – Unified compiler framework for bit-serial Processing-In-Memory architectures. Automates mapping of general-purpose workloads onto digital DRAM PIM devices. Published in *IEEE Computer Architecture Letters*, 2025.
- **PIMeval / PIMbench** – Architectural modeling and benchmarking infrastructure for digital DRAM PIM. Enables performance and energy evaluation across PIM architectures and workloads. Published in *IEEE IISWC*, 2024.

LANGUAGES

Persian (native) • English (professional working proficiency)

REFERENCES

References available upon request.